2.4 Algebraic Reasoning

Name _____ Period ____ Date _____

Geometry 2.4 (30-minute plan) · Student sheet

I can…

- justify each step when solving an equation.
- solve a formula for a variable and justify the steps.
- use the Reflexive, Symmetric, and Transitive Properties.

Vocabulary · fill in the missing word

Reflexive Property — a = _____

Symmetric Property — if a = _____, then b = a

Transitive Property — if a = _____ = c, then a = c

Part 1 · Notes and examples (fill in each note, then work the example)

Properties of equality: Addition, Subtraction, Multiplication, and Division: do the same thing to _____.

Example 1

Solve 3x + 8 = 23. Justify each step.

Distributive Property: a(b + c) = _____. **Substitution Property:** if a = b, then a can replace _____ in any equation or expression.

Example 2

Solve −2(x + 3) = 4x + 18. Justify each step.

Reflexive, Symmetric, and Transitive also apply to segment lengths and angle _____.

Example 3

Name the property.

a. m∠A = m∠A

b. If AB = CD, then CD = AB.

c. If m∠1 = m∠2 and m∠2 = m∠3, then m∠1 = m∠3.

Part 2 · Partner practice

Partner practice

Name the property: If JK = KL and KL = 12, then JK = 12.